Somatic CAG expansion in Huntington's disease: a re-analysis of released data

Craig W. French, Princeton, New Jersey · cf3357@princeton.edu · ORCID 0009-0001-8847-3018

Two groups published large studies of Huntington's disease in *Cell*, one on the striatum in 2025 and one on the cortex in 2026, and each released its underlying measurements. I re-analyzed those public files. Nothing here involved new tissue or new experiments. The figures are in `figures/Repeat_Disease_Thresholds.pdf`, seven boards, one page each, and the same content as a single web page in `figures/Repeat_Disease_Thresholds.html`.

The disease, in the terms this work uses

Huntington's disease is caused by a stretch of DNA in which the three letters CAG repeat over and over. Everyone has this repeat; people who inherit about 40 or more copies develop the disease. The repeat is unstable in the body's own cells: over a lifetime, in individual neurons, it grows. A person born with 42 copies can have neurons whose repeat has reached several hundred by the time they die.

Both papers reach the same conclusion about what that growth does. Below roughly 150 copies inside a single cell, that cell looks ordinary. Above 150 it begins to change, and the changes come in stages — first a gradual loss of the genes that make a neuron that particular kind of neuron, and later a sudden switching-on of genes a mature neuron should keep shut. The studies label these stages, and the labels are used throughout the figures: phase C for the gradual loss, phase D for the later crisis.

What each board shows

1. Where 150 sits among other repeat diseases

Around fifty diseases are caused by repeated DNA, and the table draws the 47 with published ranges on one scale, with the 150 figure marked across them. Their thresholds run from 12 copies to 680. The pattern that emerges is about location: repeats inside the part of a gene that becomes protein cause disease at short lengths, from 12 to 55 copies, while repeats outside that part need 30 to 680. Huntington's

belongs to the first group at 36 copies inherited, which makes the 150 figure a different kind of number — a count inside one cell decades later, where the 36 is what a person inherits.

2. What other repeat diseases suggest about where a threshold comes from

Five mechanisms drawn from myotonic dystrophy, fragile X and facioscapulohumeral dystrophy, each one a way a threshold can arise, and each one a prediction that could be tested in Huntington's disease. None has been. One example: in myotonic dystrophy the repeat produces RNA that soaks up a protein the cell has a limited supply of, and symptoms begin when the supply runs short. Adding more of that protein rescued model mice, which is what makes it a threshold rather than a slope.

3. The distribution of thresholds across those diseases

The shortest disease-causing length for each of the 47 diseases, counted into bins. Two groups show up, separated by a thin stretch between 100 and 199 copies where a single disease falls. Repeats in the protein-coding sequence run from 12 to 55 with a median of 27; those outside it run from 30 to 680 with a median near 82, and the diseases that work by shutting a gene down cluster near 200.

4. The re-analysis of the two papers' own data

Four pieces of work on the released measurements. I fitted, for every gene, the repeat length at which that gene starts to change, and found the striatum and the cortex behave differently: striatal genes mostly start between 100 and 170 copies, while 71 cortical genes have no start point at all and drift steadily from the shortest lengths measured. I also tested a possibility the cortex paper raises and leaves open: that the phase D crisis is the failure of a silencing system called PRC2. The test checks its gene list against two published lists of genes that PRC2 controls. The overlap is large and specific to phase D.

The board ends with a closer look at which phase D genes sit next to each other along the chromosome. HOX genes set out the body plan in the embryo and are held shut for the rest of life, and they are unusual in sitting in four tight groups rather than scattered about the genome. Of the 23 genes named HOXA to HOXD with a paralog number that appear in the cortex statistics, 11 are phase D genes; chance would put about a fifth of one gene there. Polycomb, the system that holds them shut, works on whole stretches of chromosome rather than one gene at a time, so a failure of the silencing should appear as runs of neighbors coming on together, and HOX is the one place where polycomb's targets lie side by side. Everywhere else in the genome they are far apart,

which is why the remaining phase D genes sit no closer together than chance would put them. The only other adjacent pair that comes on together is MEG3 with MEG8 on chromosome 14, at a spot held shut by DNA methylation instead.

5. How fast the repeat grows, and what slowing it would buy

The donor records include each person's age at death and the repeat length they inherited, which looks like enough to calculate a growth rate. It is not, and the reason is worth stating plainly. Age at death in this group is set by the disease. The donors who lived longest are the ones whose repeats grew slowest, so a long stretch of time and a fast expansion never occur together, and that is exactly the pairing a rate calculation needs. Across the 54 donors, inherited length and age at death correlate at −0.79. Within the 11 donors with per-neuron measurements, inherited length predicts how far the repeat traveled while age does not predict it at all.

What can be calculated without knowing the rate is how much slowing would be worth. In any growth process, speed and time enter only as a product, so cutting the speed by a given fraction stretches the remaining time by a fixed amount, whatever the underlying law. Anchored on the observed age of onset, a person born with 42 repeats expects onset at 54. Treatment that slows expansion by 29% beginning at age 30 buys ten years. The same ten years needs 42% at age 40 and 71% at age 50. Starting earlier is worth more than acting harder.

6. FAN1, the gene that ranks first across the 11 cortex donors

This board is the part that goes beyond the published work, and it is a single association in a small group of people rather than a result.

When DNA containing a repeat is copied, the repeat can slip, leaving a loop of extra CAGs bulging out of the strand. Two sets of enzymes act on that loop and pull in opposite directions. One, MutSβ, leads to the loop being written into the sequence, so the repeat ends up longer. The other, FAN1, cuts the loop out, so the repeat returns to the length it started at. Which way a cell drifts depends on the balance between them.

The released data give, for each donor, both how far that donor's repeats traveled and how much of every gene their neurons were making. I asked whether the two line up. They do, for FAN1, and in the direction the biology predicts: the donors whose repeats went furthest have the least FAN1. To see whether that means anything, I put all 17,516 genes through the identical calculation. FAN1 ranks first. No other gene in the genome matches the ordering as closely, in either direction, and none of the 1,935 genes expressed at FAN1's own level comes near it.

This leaves open which way the arrow runs. If the growth of the repeat, or the damage that follows it, lowers FAN1 in the neurons furthest along, the same pattern appears with cause and effect reversed. Tissue sampled after death cannot separate the two. The test that could would measure FAN1 before the expansion has happened — in presymptomatic carriers, or in people matched on the length they inherited who differ in when symptoms began.

7. When the silencing fails, and what sets the timing

The last board follows one idea: score each neuron by the repeat it holds rather than by the person it came from. That turns 11 donors into 78,855 individual measurements and makes four questions answerable.

The first is where along the repeat the HOX genes come on. Those genes build the body plan in the embryo and are held shut for the rest of a life, by a system called polycomb. Measuring each of the four clusters against its own level below 100 copies, all four run fifty times that level or more past 200 copies, and the uncertainty on each is far clear of no change. Four separately located groups of genes lifting together is what one shared switch failing would produce, and past 200 copies it has plainly happened.

How far back toward 150 the failure reaches is a harder question than the chart's shape suggests, and the reason is worth stating. A nucleus is not an independent observation: the eleven donors differ from one another far more than two neurons from the same donor do, so counting transcripts across a thousand nuclei as though each were a separate draw makes any effect look more certain than it is. Resampling donors, and then neurons within each donor, gives intervals that account for this. Across 140 to 199 copies, which is the window containing the 150 the original papers identify, two of the four clusters survive that treatment: HOXA at 3.7 times its baseline and HOXD at 6.5, each drawing transcripts from 6 of the 10 donors with neurons in the window. HOXC at 5.3 times rests on 3 donors and falls a little short. HOXB rests on 9 transcripts from a single donor, which is no evidence at all.

The second is whether that failure keeps step with the earlier loss of identity. It does. Take neurons with the same number of repeats, and the ones that have lost more of what makes them their kind of neuron are the ones whose HOX genes have opened. The released data include a set of control genes that should show nothing, and they show nothing, which is what rules out the comparison being driven by differences in data quality between cells. So the two stages advance together, though a correlation of this kind leaves open whether one process drives both.

The third returns to FAN1. If a neuron's own repair balance sets how fast its repeat grows, then cell types with more MutSβ relative to FAN1 should expand faster. Comparing cell types inside a single person, measuring the balance only in neurons whose repeats had barely moved, that is not what happens — the relationship is absent, and where it is strongest it runs opposite to the prediction. The correlation between people on board 6 is real, and it survives without different doses of these enzymes in individual cells behind it. The design limits what this can rule out: restricting the measurement to barely moved neurons takes a different slice of each cell type, since in a fast-expanding type those neurons are its stragglers, and cell types differ in so much else that the comparison credits these two enzymes with whatever the other differences do.

The fourth compares the two brain regions, across the 42 HOX genes present in both sets of measurements. They switch on the same genes, closely enough that the gene-by-gene agreement is near perfect. They do it at different lengths. A cortical neuron opens its HOX clusters as soon as it passes about 150 copies; a striatal neuron at the same length has not begun. The striatum is where repeats travel fastest and where the disease does its damage, so the expectation would run the other way. One reading of that, and a reading rather than a result: whatever decides the threshold belongs to the cell type rather than to the repeat.

What is in the repository

Every figure is built by a script from the released files, so each number on the boards can be traced back to the data it came from. `analysis/` holds the statistics, `notebooks/` the three Colab notebooks that produce the per-gene fits, and `figures/src/` the drawing code. `analysis/expression/` holds the FAN1 work and the reader that streams the expression matrices one block at a time, because the cortex file expands to 25 gigabytes if loaded whole; `analysis/silencing_tests.py` runs the four tests on the last board.

The input files are not included. `data/PUT_INPUT_FILES_HERE.txt` lists what each script needs and where to download it.

Running it

Everything under `analysis/` and `figures/src/` is Python and runs on an ordinary machine. `pip install -r requirements.txt` gets the four packages it needs. That holds because of how the expression matrices are read: the cortical one is a dense R

matrix of 39,560 genes by 78,855 nuclei, 1.3 gigabytes compressed and around 12 in memory, so `analysis/expression/rds_stream.py` takes it one column at a time and never holds more than a column at once.

The three notebooks are the exception. They run on an R kernel rather than Python, install their own packages in the first cell, and download the 2.9 gigabyte archive from Zenodo. They load each matrix whole, so the first notebook needs more memory than a laptop has for the cortical region and was run on a Colab high-memory runtime. The caudate matrix is 148 megabytes and runs anywhere.

Building the single web page needs Node alongside Python. Rendering the PDF also needs Playwright with a browser, and the build script stops with that instruction if it is absent.

Sources

- Kashin, Lee et al., *Cell* (2026), doi:10.1016/j.cell.2026.10.001. Zenodo record 18202664.
- Handsaker et al., *Cell* 188:623 (2025).
- von Schimmelmann et al., *Nature Neuroscience* 19:1321 (2016), supplementary tables 1 and 3.
- Depienne and Mandel, *American Journal of Human Genetics* 108:764 (2021), for the disease table.
- O'Reilly et al., *Molecular Therapy* 31:1661 (2023), for MSH3 silencing in mice.
- Senoussi et al., *Nature Communications* 16:9445 (2025), for the repair mechanism.